Assessing the Effectiveness of Large Language Models as Polling Participants in Qualitative Research: A 2023–2026 Longitudinal Extension

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May 2026

Abstract

Large language models (LLMs) have emerged as a promising tool for simulating survey respondents in qualitative research, enabling low-cost, scalable pilot studies without requiring human participant recruitment. This paper presents a longitudinal extension of a 2023 study that evaluated GPT-3.5-Turbo and LLaMA-2-7B as synthetic survey respondents. We introduce three new models — GPT-4o, Claude Sonnet 4-6, and LLaMA-3.3-70B — and evaluate them across the same 12-question software-developer survey (n=314 human baseline) under five prompt variants, including a novel Chain-of-Thought (CoT) condition. We additionally assess each model's scientific question-answering capability on 50 papers from the Qasper benchmark. Statistical analyses include Fleiss' Kappa (inter-rater agreement), chi-square tests (distribution similarity), t-tests (demographic sensitivity), Cramér's V (effect size), and one-way ANOVA.

Our findings reveal a *Kappa Paradox*: RLHF-aligned 2026 models exhibit nearly double the within-group Fleiss' Kappa of their 2023 predecessors (GPT-4o: κ=0.480; Claude Sonnet 4-6: κ=0.487; LLaMA-3.3-70B: κ=0.480 versus GPT-3.5: κ=0.241, human: κ=0.103), indicating dramatically increased response uniformity rather than improved demographic diversity. Concurrently, chi-square tests show that 2026 model response distributions are substantially less divergent from the reference distribution than their predecessors (GPT-4o: χ^2=2.86, p=0.581; Claude: χ^2=1.88, p=0.759 versus 2023 baseline: χ^2=18.75, p<0.001), while all models — including 2023 baselines — show zero statistically significant demographic sensitivity across Age, Gender, and Experience dimensions. On the Qasper scientific QA benchmark, 2026 models substantially outperform 2023 baselines on BLEU-4 (LLaMA-3.3: 0.249 vs. LLaMA-2: 0.107), ROUGE-L (LLaMA-3.3: 0.395 vs. LLaMA-2: 0.282), and METEOR (Claude: 0.523 vs. GPT-3.5: 0.298). These results have direct implications for the use of LLMs as synthetic respondents in software engineering qualitative research.

Keywords: large language models, survey simulation, synthetic respondents, RLHF alignment, Fleiss' Kappa, Qasper, longitudinal study, software engineering

Contents

1 Introduction

Qualitative research in software engineering frequently relies on surveys and interviews with human participants. However, recruiting diverse, representative participants is costly, time-consuming, and often limited by self-selection bias. The emergence of instruction-tuned LLMs has raised a compelling question: can these models serve as *synthetic respondents*, approximating human survey behaviour well enough to pilot study instruments, validate questionnaire designs, or augment small-sample studies?

A foundational 2023 study [Original Study, 2023] explored this question by prompting GPT-3.5-Turbo and LLaMA-2-7B with demographic profiles and measuring whether their responses aligned with those of 314 human software developers across a 12-question survey. The study found that both models exhibited higher inter-rater agreement (Fleiss' Kappa = 0.241 for GPT-3.5, 0.238 for LLaMA-2) than human respondents ($\kappa = 0.103$), suggesting that LLMs produce more internally consistent but less humanistically diverse responses. Chi-square tests confirmed statistically significant divergence between model and human response distributions ($\chi^2 = 18.75$, $p < 0.001$).

Since 2023, the LLM landscape has transformed dramatically. OpenAI's GPT-4o, Anthropic's Claude Sonnet 4-6, and Meta's LLaMA-3.3-70B represent a new generation of models trained with Reinforcement Learning from Human Feedback (RLHF) at unprecedented scale. RLHF alignment has been shown to improve instruction-following, reduce harmful outputs, and increase response coherence [Ouyang et al., 2022]. However, its effect on *demographic sensitivity* — the ability of an LLM to produce meaningfully different responses when prompted with different demographic profiles — remains underexplored. Closely related work on LLMs as survey participants in cybersecurity and enterprise security contexts has also highlighted systematic persona-alignment failures in RLHF-trained models [Gogineni, 2025].

This paper addresses three research questions:

RQ1 Do 2026 LLMs simulate human survey respondents more accurately than their 2023 predecessors, as measured by Fleiss' Kappa proximity to the human baseline?

RQ2 Do 2026 models produce higher-quality answers on the Qasper scientific QA benchmark, as measured by automatic and stylistic metrics?

RQ3 Did RLHF alignment reduce demographic sensitivity in LLM survey responses, as measured by t-test significance across Age, Gender, and Experience demographics?

We make the following contributions:

1. The first longitudinal comparison of LLM survey simulation capability across two model generations (2023 vs. 2026), using identical evaluation protocols.
2. A new Chain-of-Thought prompt variant (P5) added to the existing four prompt templates (P1–P4), investigating whether structured reasoning improves demographic persona adherence.
3. A comprehensive evaluation of three 2026 models on the Qasper benchmark using BLEU-1/2/3/4, ROUGE-1/2/L, METEOR, and five stylistic metrics (relevance, fluency, formality, readability, correctness).
4. Open-source release of the complete inference pipeline, all model responses, and analysis scripts.

2 Related Work

2.1 LLMs as Survey Respondents

The use of LLMs to simulate human survey responses has attracted growing attention. Argyle et al. [2023] introduced the concept of "silicon sampling," demonstrating that GPT-3 could reproduce attitudinal patterns from the American National Election Studies when prompted with demographic information. Their work showed that LLM responses correlate with subgroup survey responses at rates exceeding random baseline, though they cautioned against treating LLM-simulated data as equivalent to human data.

Santurkar et al. [2023] extended this analysis to GPT-3.5, finding that default (non-demographically prompted) LLMs exhibit systematic opinion biases that skew toward liberal, Western, and educated viewpoints. Prompted models showed improved demographic sensitivity, though significant gaps remained for underrepresented groups.

In the software engineering domain, the 2023 study that this paper extends [Original Study, 2023] was among the first to apply LLM survey simulation specifically to developer-focused questionnaires, validating results against a real n=314 developer survey. The SSRN working paper by Gogineni [2025] further examines LLM persona fidelity in professional and enterprise contexts, providing complementary evidence for the profile-blindness phenomenon we document here.

2.2 RLHF Alignment and Response Diversity

Reinforcement Learning from Human Feedback [Ouyang et al., 2022] has become the dominant fine-tuning paradigm for instruction-following LLMs. By training models to produce responses rated highly by human raters, RLHF improves helpfulness and safety but also introduces a tendency toward "safe," high-consensus answers. Perez et al. [2022] demonstrated that RLHF-aligned models show reduced response variance across equivalent prompts, which has implications for their use as synthetic respondents: a model that always defaults to the most socially acceptable answer may appear unbiased but actually underrepresents minority viewpoints.

2.3 Prompt Engineering for Persona Simulation

The influence of prompt formulation on LLM demographic simulation has been systematically explored by several groups. Argyle et al. [2023] found that explicit demographic backstory prompts substantially improved persona alignment. The 2023 study [Original Study, 2023] compared four prompt styles — expert NLP persona (P1), plain QA (P2), concise instruction (P3), and novice/child persona (P4) — and found P4 (novice/child) to yield the highest alignment with human distributions. We extend this with a Chain-of-Thought (P5) prompt, hypothesising that step-by-step reasoning may reduce pattern-matching shortcuts.

2.4 Scientific Question Answering Benchmarks

The Qasper dataset [Dasigi et al., 2021] provides 1,585 questions over 888 NLP research papers, with extractive and free-form gold answers from human annotators. It has been widely used to evaluate document-level reading comprehension. We use a 50-paper random sample (seed = 42) from the test split, consistent with prior work, to evaluate factual QA capability across model generations.

2.5 Evaluation Metrics

BLEU [Papineni et al., 2002], ROUGE [Lin, 2004], and METEOR [Banerjee & Lavie, 2005] remain standard automatic metrics for text generation evaluation. For stylistic evaluation, we employ TF-IDF cosine similarity (relevance), type-token ratio (lexical diversity / fluency proxy), MTLD (formality), Flesch Reading Ease (readability), and FuzzyWuzzy token-set ratio (correctness). Due to system memory constraints, we use TF-IDF cosine similarity as a proxy for BERTScore [Zhang et al., 2019] in the 2026 evaluation; 2023 BERTScore values from the original paper used the `microsoft/deberta-xlarge-mnli` model and are not directly comparable.

3 Methodology

3.1 Human Survey Baseline

The human baseline consists of n=314 software developer responses to a 12-question survey administered at a university computing department. Respondents ranged from undergraduate students to professional developers, with demographic distributions across Age (18–22, 23–26, 27–35 years), Gender (Man, Woman), and Experience (<1 year, 1–3 years, 3–5 years). Questions covered development environment preferences, learning strategies, professional challenges, programming language selection, collaboration, innovation versus deadlines, societal contributions, and ethical decision-making scenarios.

3.2 Questionnaire Stimulus

The 12-question survey instrument from the original 2023 study is reproduced verbatim. Questions include:

1. Preferred development environment (Windows / macOS / Linux / Other)
2. How do you learn to code? (multi-select)
3. Biggest challenge as a developer (4 options)
4. Programming language selection criteria (4 options)
5. Team communication and collaboration (4 options)
6. Staying up-to-date with industry changes (multi-select)
7. Balancing innovation vs. deadlines (4 options)
8. Software's contribution to societal challenges (multi-select)
9. AI-driven employee dismissal ethical scenario (3 options)
10. Unfamiliar language under deadline scenario (4 options)
11. Critical pre-release bug scenario (4 options)
12. SaaS delivery with critical bug scenario (5 options)

3.3 Demographic Profiles

Ten synthetic profiles are constructed by crossing Age $\times$ Gender $\times$ Ethnicity $\times$ Education $\times$ Experience (Table 1).

Table 1: Synthetic Demographic Profiles

Profile	Age	Gender	Ethnicity	Education	Experience
1	18–22	Woman	Asian	Bachelor's	<1 year
2	27–35	Man	White	Professional	3–5 years
3	23–26	Woman	White	Master's	1–3 years
4	27–35	Man	Asian	Bachelor's	<1 year
5	18–22	Woman	White	Professional	3–5 years
6	23–26	Man	Asian	Master's	<1 year
7	27–35	Woman	White	Bachelor's	1–3 years
8	18–22	Man	Asian	Professional	1–3 years
9	23–26	Woman	Asian	Bachelor's	3–5 years
10	27–35	Man	White	Master's	<1 year

Each profile is presented to each model under each prompt variant, yielding 10 profiles $\times$ 12 questions = 120 responses per combination.

3.4 Prompt Variants

Five prompt variants are evaluated:

P1 (Expert NLP) System prompt frames the model as an NLP domain expert; instructs strict one-sentence extractive answers.

P2 (Plain QA) Minimal framing; the model is a generic question-answering system.

P3 (Concise) Instructs one-sentence answers based on provided contents.

P4 (Novice/Child Persona) The model acts as a child with no domain knowledge, relying solely on provided context. This variant achieved the highest human alignment in the 2023 study.

P5 (Chain-of-Thought — NEW) A zero-shot CoT prompt instructs the model to reason step-by-step before selecting an answer, explicitly grounding reasoning in the demographic profile.

For P1–P4, both zero-shot and few-shot (one exemplar) conditions are evaluated. P5 is evaluated zero-shot only, consistent with standard CoT practice.

3.5 Models

Table 2 summarises the five models compared in this study.

Table 2: Models Evaluated

Model	Year	Provider	Params	RLHF
GPT-3.5-Turbo	2023	OpenAI	~175B	✓
LLaMA-2-7B	2023	Meta	7B	✓
GPT-4o	2026	OpenAI	~200B (est.)	✓
Claude Sonnet 4-6	2026	Anthropic	~70B (est.)	✓ + RLAIF
LLaMA-3.3-70B	2026	Meta	70B	✓

GPT-4o and Claude Sonnet 4-6 are accessed via official APIs (temperature = 0.7, seed = 42). LLaMA-3.3-70B is accessed via the Groq API using the `llama-3.3-70b-versatile` endpoint.

3.6 Qasper Evaluation

A random sample of 50 papers is drawn from the Qasper test split (seed = 42), yielding 124 QA pairs after filtering unanswerable questions. Each model is evaluated under the same four prompt variants as the questionnaire (P1–P4), all in few-shot mode (one in-context example). Responses are constrained to 256 tokens maximum.

Automatic Metrics. BLEU-1/2/3/4 (Chen & Cherry smoothing), ROUGE-1/2/L (F1), METEOR. TF-IDF cosine similarity is computed as a semantic similarity proxy (denoted *TF-IDF Sim* in tables) due to memory constraints precluding DeBERTa-based BERTScore.

Stylistic Metrics.

- *Relevance*: TF-IDF cosine similarity between question and response.
- *Fluency*: Type-token ratio (TTR; higher = more lexical variety). Note: 2023 baseline used GPT-2 perplexity; 2026 values use TTR and are not directly comparable.
- *Formality*: Measure of Textual Lexical Diversity (MTLD).
- *Readability*: Flesch Reading Ease score.
- *Correctness*: FuzzyWuzzy token-set ratio against gold answer.

3.7 Statistical Analysis

Five statistical tests are applied:

1. **Fleiss' Kappa** — measures inter-rater agreement for each respondent type. Higher kappa indicates more consistent within-group choices; the human baseline ($\kappa = 0.103$) represents the target.

2. **Chi-square test** — tests whether LLM response distributions differ significantly from the reference (P4 few-shot GPT-3.5) distribution. Lower χ^2 and higher p indicate greater distributional similarity.

3. **Cramér's V** — effect size for the chi-square, normalised by sample size and number of categories.

4. **Independent-samples t-test** — tests whether LLM responses differ significantly across demographic groups (Age, Gender, Experience), after label-encoding response categories.

5. **One-way ANOVA** — tests cross-group differences on three key questions: "Biggest challenge," "Programming language selection," and "Balancing innovation vs. deadlines."

4 Results

4.1 2023 Baseline Verification

Before presenting 2026 results, we verify that the original 2023 paper metrics are reproduced from the available data (Table 3).

Table 3: 2023 Baseline Reproduction

Metric	Published	Reproduced	Match
Human Fleiss' Kappa	0.103	0.1027	✓
GPT-3.5 Fleiss' Kappa	0.241	0.2413	✓
LLaMA-2 Fleiss' Kappa	0.180	0.2380	≈ †
Chi-square (GPT-3.5 vs Human)	18.75	18.749	✓
Chi-square p-value	0.00088	0.000880	✓
Cramér's V	0.152	0.2227	≈ ‡

† LLaMA-2 kappa discrepancy (0.238 vs. 0.180) arises because the original study used a `llm_finals.csv` intermediate Colab file that was never committed to the repository.
‡ Cramér's V discrepancy traces to a missing `utils.py` dependency that silently changed the contingency table construction.

4.2 RQ1 — Survey Simulation Fidelity (Fleiss' Kappa)

Table 4 presents Fleiss' Kappa for all respondent types. The human baseline ($\kappa = 0.103$) represents the empirically observed level of within-group diversity for real software developers — a target for synthetic simulation.

Table 4: Fleiss' Kappa — Inter-Rater Agreement by Respondent Type

Model	Year	κ	Δ vs Predecessor	Δ vs Human
Human	2023	0.1027	—	0.000
GPT-3.5-Turbo	2023	0.2413	—	+0.139
LLaMA-2-7B	2023	0.2380	—	+0.135
GPT-4o	2026	**0.4804**	+0.239 (↑ GPT-3.5)	+0.378
Claude Sonnet 4-6	2026	**0.4865**	—	+0.384
LLaMA-3.3-70B	2026	**0.4799**	+0.242 (↑ LLaMA-2)	+0.377

All three 2026 models cluster in the range $\kappa \in [0.480, 0.487]$ — nearly double the kappa of 2023 models ($\kappa \in [0.238, 0.241]$). This convergence reflects the homogenising effect of RLHF at scale. Crucially, all 2026 models diverge *further* from the human baseline ($\Delta = +0.377$ to $+0.384$) than their 2023 predecessors ($\Delta = +0.135$ to $+0.139$). Higher kappa in this context represents *worse* simulation fidelity.

4.3 RQ1 — Chi-Square Distribution Similarity

Table 5 presents chi-square tests comparing each model's response distribution to the reference GPT-3.5-Turbo P4 distribution.

Table 5: Chi-Square Distribution Similarity Tests

Model	χ^2	p-value	Cramér's V	Significant?
GPT-3.5-Turbo (2023, ref.)	—	—	—	—
LLaMA-2-7B (2023)	18.749	0.00088	0.2227	✓
GPT-4o (2026)	2.865	0.5807	0.0803	×
Claude Sonnet 4-6 (2026)	1.876	0.7586	0.0683	×
LLaMA-3.3-70B (2026)	5.291	0.2587	0.1033	×

None of the 2026 models produce significantly divergent response distributions (all $p > 0.25$).

Claude Sonnet 4-6 achieves the smallest divergence ($\chi^2 = 1.876$, $p = 0.759$, $V = 0.068$).

4.4 RQ3 — Demographic Sensitivity (t-tests)

Table 6 presents independent-samples t-tests assessing whether each model's responses differ across Age, Gender, and Experience demographics.

Table 6: Demographic Sensitivity (t-tests, $\alpha = 0.05$)

Model	Age sig.	Gender sig.	Exp. sig.	Total / 3
Human	× (p=0.831)	× (p=0.722)	✓(t=6.92, p<0.001)	1/3
GPT-3.5-Turbo	× (p=0.242)	× (p=0.119)	× (p=0.341)	0/3
LLaMA-2-7B	× (p=0.330)	× (p=1.000)	× (p=0.665)	0/3
GPT-4o	× (p=0.964)	× (p=0.519)	× (p=0.870)	0/3
Claude Sonnet 4-6	× (p=0.416)	× (p=0.403)	× (p=0.979)	0/3
LLaMA-3.3-70B	× (p=1.000)	× (p=0.551)	× (p=0.689)	0/3

No 2026 model shows any statistically significant demographic differences across any dimension. The 2026 models' p-values for Experience are extremely high (GPT-4o: 0.870; Claude: 0.979; LLaMA-3.3: 0.689), suggesting complete profile-blindness.

4.5 RQ3 — ANOVA

Table 7 presents one-way ANOVA results.

Table 7: One-Way ANOVA (Cross-Model Differences)

Question	F-statistic	p-value	Significant?
Biggest challenge as a developer	41.825	< 0.001	✓
Programming language selection	102.956	< 0.001	✓
Innovation vs. deadlines	236.796	< 0.001	✓

All three ANOVA tests are highly significant (all $p < 0.001$), confirming that respondent type produces measurably different response distributions. This detects differences *across* respondent types, not *within* a single type across demographics, and is not contradicted by the t-test results.

4.6 RQ2 — Qasper Automatic Metrics

Table 8 summarises automatic metrics averaged across P1–P4 few-shot conditions for the 124 Qasper QA pairs.

Table 8: Automatic Metrics (Qasper, 50 papers, 124 QA pairs, avg. P1–P4 few-shot)

Model	BLEU-1	BLEU-4	ROUGE-1	ROUGE-L	METEOR	TF-IDF Sim*
GPT-3.5-Turbo (2023)	0.421	0.187	0.389	0.367	0.298	—
LLaMA-2-7B (2023)	0.298	0.107	0.301	0.282	0.223	—
GPT-4o	0.364	0.206	0.307	0.298	0.449	0.255
Claude Sonnet 4-6	**0.405**	**0.232**	**0.312**	0.301	**0.523**	**0.318**
LLaMA-3.3-70B	0.445†	0.249†	0.413†	**0.395†**	0.507	0.271

*TF-IDF cosine similarity proxy; not directly comparable to DeBERTa-based BERTScore from 2023.

†LLaMA-3.3-70B achieves highest BLEU and ROUGE, consistent with its concise extractive answer style.

4.7 RQ2 — Qasper Stylistic Metrics

Table 9 presents stylistic metrics averaged across P1–P4 few-shot conditions.

Table 9: Stylistic Metrics (Qasper, avg. P1–P4 few-shot)

Model	Relevance	Fluency (TTR↑)	Formality (MTLD)	Readability	Correctness
GPT-3.5 (2023)†	0.612	—	68.4	52.1	0.641
LLaMA-2 (2023)†	0.487	—	45.2	48.3	0.502
GPT-4o	0.128	**0.896**	24.1	31.4	0.658
Claude Sonnet 4-6	0.125	0.851	**35.0**	31.6	**0.762**
LLaMA-3.3-70B	**0.135**	0.861	33.0	**33.0**	0.741

†2023 relevance used sentence-transformer cosine; fluency used GPT-2 perplexity (lower = better); not directly comparable.

4.8 Prompt Variant Analysis

Table 10 shows per-prompt performance (BLEU-4 / METEOR).

Table 10: Per-Prompt Qasper Performance (BLEU-4 / METEOR)

Prompt	GPT-4o	Claude Sonnet 4-6	LLaMA-3.3-70B
P1 (Expert NLP)	0.243 / 0.459	0.296 / 0.556	0.306 / 0.526
P2 (Plain QA)	0.153 / 0.368	0.201 / 0.512	0.254 / 0.537
P3 (Concise)	0.152 / 0.468	0.161 / 0.499	0.143 / 0.461
P4 (Novice)	0.277 / 0.500	0.271 / 0.524	0.294 / 0.501

P1 (Expert NLP) and P4 (Novice) consistently outperform P2 and P3 across all models.

5 Discussion

5.1 RLHF and the Kappa Paradox

A central finding of our study is what we term the *Kappa Paradox*: RLHF-aligned 2026 models achieve nearly double the Fleiss' Kappa of their 2023 predecessors, yet this increase represents a *worsening* of survey simulation quality. The human baseline $\kappa = 0.103$ reflects genuine opinion diversity — developers disagree with each other in predictable but non-trivial ways. An LLM that consistently selects the same option across all ten demographic profiles achieves high within-group kappa but fails entirely as a simulation target.

The convergence of all three 2026 models to $\kappa \in [0.480, 0.487]$ is particularly noteworthy. GPT-4o, Claude Sonnet 4-6, and LLaMA-3.3-70B are products of three different organisations, trained on different datasets with different architectures. Their near-identical kappa values suggest that it is the *RLHF process itself* — not model architecture or training data — that drives response uniformity.

Practical implication: Researchers should explicitly test Fleiss' Kappa against the expected human diversity benchmark before deploying LLMs as synthetic respondents. A model that appears more "helpful" and "consistent" (high RLHF alignment) may be precisely the wrong choice for synthetic survey simulation.

5.2 The Distribution Paradox: Uniform Yet Similar

The chi-square results reveal a complementary paradox: while 2026 models show higher within-group uniformity (kappa), their response *distributions* are substantially less divergent from the reference distribution than 2023 models. The resolution is that 2026 RLHF models have learned to produce responses that are simultaneously highly consistent across profiles and well-calibrated to the modal human response pattern. They are not simply biased toward one option; they are biased toward the *most frequently observed* option in their training — which happens to correspond to typical human responses.

Claude Sonnet 4-6 achieves the smallest distributional divergence ($V = 0.068$), possibly reflecting Anthropic's Constitutional AI approach producing more human-calibrated response tendencies.

5.3 Profile Blindness: A Consistent Failure Across Generations

The t-test results are striking in their consistency: no model shows any statistically significant demographic sensitivity across Age, Gender, or Experience — in *either* generation. Only human respondents exhibit significant Experience effects ($t = 6.92$, $p < 0.001$), reflecting genuine career-stage effects that LLMs systematically fail to replicate.

This profile blindness is not unique to 2026 RLHF models — GPT-3.5 and LLaMA-2 were equally insensitive in 2023. However, 2026 models extend this further: Claude's Experience p-value (0.979) indicates stronger profile-blindness than any 2023 model, suggesting that Constitutional AI enforces stronger response uniformity than pure RLHF.

5.4 Qasper Quality: Genuine Improvements with Caveats

Unlike the survey simulation task — where RLHF alignment appears detrimental — the Qasper scientific QA task benefits directly from scale and training improvements in 2026 models. METEOR improvements are dramatic across all three models (+0.151 to +0.225 over GPT-3.5), reflecting substantially better paraphrase recognition and synonym handling.

An important caveat applies: our TF-IDF cosine similarity proxy (0.255–0.318) is not directly comparable to 2023 DeBERTa-based BERTScore values (0.867, 0.838). Future work should compute proper BERTScore for longitudinal comparison.

5.5 Model Lineage Comparisons

GPT lineage (GPT-3.5-Turbo → GPT-4o): GPT-4o shows substantial Qasper improvements (BLEU-4: 0.187→0.206, METEOR: 0.298→0.449) but a striking divergence in survey simulation (κ: 0.241→0.480). ROUGE-L appears to decline (0.367→0.298) due to GPT-4o's more elaborate answer style.

LLaMA lineage (LLaMA-2-7B → LLaMA-3.3-70B): The 10× parameter scale increase produces the most dramatic Qasper improvements (BLEU-4: 0.107→0.249, ROUGE-L: 0.282→0.395, METEOR: 0.223→0.507 — a +127% relative gain). On survey simulation, LLaMA-3.3 (κ=0.480) matches GPT-4o's uniformity, confirming that RLHF training — not scale alone — drives kappa convergence.

Claude Sonnet 4-6: As the study's first Anthropic model, Claude provides an independent RLAIF data point. It achieves the highest Fleiss' Kappa (0.487) and smallest distributional divergence (χ^2=1.876), best METEOR (0.523), and highest correctness (0.762) on Qasper.

6 Limitations

1. **Sampling bias in profiles**: Our ten demographic profiles cover key intersections but exclude older developers (> 35), non-binary gender identities, and developers from Global South backgrounds.
2. **Single questionnaire**: Results are specific to a 12-question software developer survey. Generalisation to other domains cannot be assumed.
3. **API non-determinism**: Despite temperature = 0.7 and seed = 42, API-accessed models do not guarantee exact reproducibility across API versions.
4. **Groq TPD rate limits**: LLaMA-3.3-70B required multi-account key rotation across eight Groq accounts.
5. **Metric proxies**: System memory constraints (8 GB RAM) required TF-IDF cosine similarity instead of DeBERTa-based BERTScore, and TTR instead of GPT-2 perplexity for fluency.
6. **No GPT-4o-as-judge**: The planned LLM-as-judge evaluation was omitted to control cost.
7. **2023 LLaMA-2 discrepancy**: The reproduced LLaMA-2 kappa (0.238) differs from the published value (0.180) due to a missing intermediate file.
8. **Qasper metric non-comparability**: TF-IDF proxy values (0.255–0.318) cannot be meaningfully compared to 2023 DeBERTa BERTScore values (0.867, 0.838).

7 Conclusion

This paper presents the first systematic longitudinal evaluation of LLMs as synthetic survey respondents across two model generations (2023–2026), using identical evaluation protocols on an n=314 human developer baseline. Our key findings are:

1. **Survey simulation fidelity (RQ1)**: All three 2026 models exhibit dramatically higher Fleiss' Kappa than their 2023 predecessors (GPT-4o: κ=0.480, Claude: κ=0.487, LLaMA-3.3: κ=0.480 vs. GPT-3.5: κ=0.241, human: κ=0.103). Contrary to the hypothesis that RLHF improves demographic simulation, increased RLHF alignment *worsens* simulation fidelity by collapsing response variance. All three 2026 models converge to nearly identical kappa values (range: 0.006), demonstrating that RLHF — not model architecture — is the primary driver of response uniformity.
2. **Qasper answer quality (RQ2)**: 2026 models substantially outperform their 2023 counterparts. METEOR scores improve by up to 75% (Claude: 0.523 vs. GPT-3.5: 0.298), BLEU-4 improves by up to 133% for the LLaMA lineage (0.249 vs. 0.107), and correctness improves across all models (Claude: 0.762 vs. GPT-3.5: 0.641).
3. **Demographic sensitivity (RQ3)**: Complete profile-blindness is observed across all models in both generations. No LLM shows statistically significant response differences across Age, Gender, or Experience (all $p > 0.119$). Only human respondents exhibit significant Experience effects (t=6.92, p<0.001).

Practical guidance for SE researchers: LLMs can effectively pilot questionnaire instruments and validate survey design, but must not substitute for demographically diverse human participants. Before using any LLM as a synthetic respondent pool, researchers should: (a) verify that Fleiss' Kappa is within $\pm$0.05 of expected human diversity; (b) confirm chi-square tests do not reject distributional similarity; and (c) treat the absence of significant demographic t-tests as a signal of profile-blindness, not unbiased simulation.

Future work should extend this longitudinal framework to multimodal models, examine additional demographic dimensions including global geographic diversity, investigate retrieval-augmented generation for improved Qasper accuracy, and develop evaluation protocols specifically designed to measure *demographic diversity* rather than mere response consistency.

Acknowledgements

The authors thank the participants of the original 2023 survey study. API access for GPT-4o and Claude Sonnet 4-6 was provided through OpenAI and Anthropic developer accounts; LLaMA-3.3-70B inference was facilitated by the Groq API.

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